

ABHASA CRM

Mental Health Care Management Platform

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete and precise description of all functional requirements, business rules, data models, workflows, and validations for the Abhasa CRM system — a dedicated Mental Health Care Management Platform. The document is generated directly from the implemented codebase and reflects only actual, implemented features.

1.2 Scope

Abhasa CRM is a multi-role React/TypeScript web application that manages the full patient acquisition and care lifecycle: from lead creation and duplicate detection, through consultation scheduling and live session management, to post-session feedback and weekly agent performance review. The system supports four distinct user roles:

- Admin — full platform access
- BDM Team Leader — team management and reporting
- BDM Team Member / Agent — lead and follow-up management
- Psychologist — consultation and session management

1.3 System Overview

The platform integrates with WATI (WhatsApp Business API), Razorpay (payment/refund processing), and PCMS (Patient Care Management System for Rehab referrals). All routes are protected by role-based access. Leads can be created by both the AI system and BDM team members.

1.4 Modules & Definitions

Modules	Definition
Lead	A caller who has contacted the organization regarding mental health services for themselves or a patient
Patient	The person requiring mental health care (may differ from caller/lead)
BDM	Business Development Manager — responsible for lead qualification and consultation booking
Psychologist	Licensed mental health professional conducting consultations
WATI	WhatsApp Business API integration for patient communication
Consultation	A scheduled or completed session between patient and psychologist
Session	The live video/audio call conducted within the Consultation Room

Modules	Definition
Follow-up	Post-lost-lead re-engagement activity assigned to agents
Lead Score	AI-computed numeric score (0–10) indicating lead quality/intent
Temperature	Qualitative lead classification: Hot, Warm, Cold
createdBy	Whether a lead was created by the AI system or a BDM team member

MODULE 1: Call Management Module

Description

The Call Management Module provides a centralised view of all inbound and outbound communication records associated with each lead. Accessible from the Leads table via the History icon button, it renders as a side Sheet displaying all calls grouped by callerName across all leads. Each call row is clickable to reveal AI-generated insights in a panel below the table, including sentiment analysis, intent detection, outcome classification, and agent coaching notes. The module also surfaces an Activity Timeline for the selected lead.

Functional Overview

- Call history is accessed by clicking the History icon button on any lead row in the Leads table.
- A Sheet slides in from the right displaying all calls linked to the same callerName across all leads.
- Calls are sourced from the calls data store and filtered by callerLeadIds.
- Calls are sorted newest-first by timestamp (descending).
- Each call row shows: Date & Time, Patient Name, Duration, Agent, Sentiment, and Outcome.
- Clicking a call row activates the AI Insights panel below the table.
- The AI Insights panel shows: Call Summary, Sentiment, Intent, Outcome, Lead Score, Improvement Notes, and Next Action.
- An Activity Timeline is displayed below the AI Insights panel for the selected lead.
- If no calls exist, an empty-state illustration is shown.

Fields Specification — Call Record

Field Name	Data Type	Required	Description
id	UUID (String) — system auto-generated	Yes	Unique identifier for the call record
leadId	UUID (String) — references Lead record	Yes	Lead ID linking this call to the patient lead
agentId	UUID (String) — references User Master (Agent role)	Yes	Agent who handled or was assigned to this call
timestamp	ISO DateTime	Yes	Date and time when the call was made or received

Field Name	Data Type	Required	Description
duration	Number (seconds)	Yes	Total call duration in seconds; displayed as Xm Ys in the UI
critical	Boolean	No	Flags calls requiring attention; renders row with destructive background highlight
aiInsights.summary	String	Yes	AI-generated narrative summary of the call content
aiInsights.sentiment	String — AI-computed: positive, negative, or neutral	Yes	Overall call sentiment; mapped to StatusChip: positive=Excellent, negative=Critical, neutral=Good
aiInsights.intent	String	Yes	Caller's primary detected intent (e.g., consultation inquiry)
aiInsights.outcome	String — AI-computed: Converted, Follow-up, Callback, or No Action	Yes	Call outcome classification computed by AI
aiInsights.leadScore	Number (0-10)	Yes	AI-computed lead quality score specific to this call
aiInsights.improvementNotes	String	Yes	AI-generated coaching note for the agent handling this call
aiInsights.nextAction	String	Yes	AI-recommended next step based on call analysis
aiInsights.agentScores.empathy	Number (0-10)	Yes	AI score for agent empathy demonstrated during this call
aiInsights.agentScores.professionalism	Number (0-10)	Yes	AI score for agent professionalism during this call
aiInsights.agentScores.conversionSkill	Number (0-10)	Yes	AI score for agent conversion effectiveness during this call

Business Rules

- callerLeadIds is computed as all lead IDs where callerName matches the selected lead — all calls across any lead belonging to the same caller are shown.
- Calls are sorted by timestamp descending (newest first).
- activeSheetCall is set when a call row is clicked; only one call can be active at a time.
- Sentiment is mapped to StatusChip labels: positive=Excellent, negative=Critical, neutral=Good.
- Critical flag renders the call row with a destructive/5 background tint.
- Duration is formatted as Xm Ys (e.g., 4m 32s) in the call table.
- Activity Timeline below the AI Insights panel shows entries from both localActivityLogs and the lead's persisted activityLog, sorted newest first.
- The Sheet closes when the user clicks outside or presses Escape, resetting activeSheetCallId to null.

Workflow

- User clicks the History icon button on any lead row in the Leads table.
- The Call History Sheet opens; historyLead is set to the selected lead.
- callerLeadIds is computed from all leads sharing the same callerName.
- leadCalls is filtered and sorted: all calls where leadId is in callerLeadIds, newest first.
- If no calls exist, an empty-state (PhoneCall icon + message) is displayed.
- User clicks a call row — activeSheetCallId is set; the AI Insights panel renders below the table.
- User views Summary, Sentiment, Intent, Outcome, Lead Score, Improvement Notes, and Next Action.
- Activity Timeline renders below AI Insights, showing all activity events for the lead.
- User closes the Sheet — historyLead and activeSheetCallId are reset to null.

Dependencies

- calls data store (imported from @/data/calls)
- useLeads() context for leads array
- agents data for agent name and avatar resolution
- ActivityTimeline component for the activity log section
- StatusChip component for sentiment and outcome display

MODULE 2: Lead Creation Module

Description

The Lead Creation Module enables both automated AI systems and BDM team members to create new patient leads within the CRM. The module captures comprehensive caller and patient information, supports multi-patient association per caller, and initiates the lead lifecycle by automatically generating a 'Lead Created' activity log entry upon successful submission.

Functional Overview

- New leads are created through the **NewLeadDialog** component from the **Leads** page.
- The system supports **two lead creation methods**:
 - **AI-generated leads** imported automatically
 - **Manual leads** entered by **BDM team members**
- Both lead creation methods go through the **same duplicate detection process**.
- The system checks for **duplicate leads before saving**.
- After verification, leads are stored using the **LeadsContext** data store.

Fields Specification

Field Name	Data Type	Required	Description
id	id (UUID)	Yes	Unique system-generated identifier for the lead record
callerName	String	Yes	Full name of the person who placed the call
patientName	String	Yes	Full name of the patient requiring care (may differ from caller)
relationshipWithPatient	id (Select Dropdown)	Yes	Caller's relationship to patient: Self Mother Father Spouse Friend
phone	String	Yes	Primary contact phone number of the caller (used for WhatsApp/WATI)
alternatePhone	String	No	Secondary contact phone number for the caller
email	String	Yes	Email address of the caller for follow-up communication
city	id (Select Dropdown)	Yes	City of residence — used in duplicate detection scoring
source	id (Select Dropdown)	Yes	Acquisition channel: Facebook Instagram Website Referral
status	id (Select Dropdown)	Yes	Lead status: New Active Converted Lost Affordable Rehab Interested Follow-up Not Interested

Field Name	Data Type	Required	Description
temperature	id (Select Dropdown)	Yes	Lead quality classification: Hot Warm Cold
duration	Number	Yes	Duration of the call
leadScore	Number (0–10)	Yes	AI-computed lead quality score from 0 to 100
assignedBDM	id (BDMID)	No	ID of the BDM supervisor overseeing this lead
psychologistAssigned	id (PsychID)	No	ID of the psychologist assigned to conduct consultations
Consultation Details	Id from consultation	No	Current consultation state: Scheduled Completed Cancelled
lastActivity	ISO DateTime	Yes	Timestamp of the most recent action on this lead
createdAt	ISO DateTime	Yes	Timestamp when the lead record was first created
createdBy	string	Yes	Origin of lead creation: AI BDM
followUpAssigned To	id (AgentID)	No	Agent assigned for post-lost follow-up activity
followUpDate	ISO DateTime	No	Scheduled date for the next follow-up call
followUpStatus	Id	No	Current follow-up state: Pending Called Not Reachable Converted Rescheduled
medicalProblem	String	No	Free-text description of the patient's medical condition
leadMedium	Id (Select Dropdown)	No	Specific acquisition medium (e.g., Google Ads, Direct Call)
enquiryType	id (Select Dropdown)	No	Nature of inbound enquiry: Promotion Jobs Collaboration Telecalling Rehab Follow-up Patient Enquiry
purposeOfCall	id (Select Dropdown)	No	Caller's stated reason: Promotion Jobs Collaboration Telecalling Rehab Follow-up Patient Enquiry General Inquiry
caseType	id (Select Dropdown)	No	Clinical classification: Addiction Custom Wellness Detox Others
serviceType	id (Select Dropdown)	No	Delivery mode: Online Session Rehab In-Person Other
notes	String (Multiline)	No	Free-text agent notes about the lead
transactionHistory	Id from Patient	No	List of all payment transactions associated with this lead
leadHistory	Array<LeadHistoryEntry>	Yes	Immutable audit trail of all lifecycle events

Field Name	Data Type	Required	Description
Call Log	Array	No	List of inbound and outbound communication records associated with the lead
activityLog	Array<Activity Entry>	No	Dynamic log of status changes, notes, follow-ups, and calls

Business Rules

- Leads created by the AI system have createdBy = 'AI'; manually created leads have createdBy = 'BDM'.
- Upon creation, the system automatically appends a 'Lead Created' activity log entry with kind='created' and the current timestamp.
- If strictAssign is true, the resolveAgent() function forces assignedAgent to equal lastAgentId, preventing any automated re-routing.
- Each caller can be associated with multiple patients. The callerName field is the grouping key for patient-level expansion in the Leads table.
- The patientOnlyIds set tracks patient-only records so they are excluded from the main leads list view.
- A lead with serviceType = 'Rehab' triggers a 'Push to PCMS' workflow banner in the LeadDetail page.
- A lead with serviceType = 'Online Session' displays a 'Book Session' workflow banner.

Validation Rules

- callerName, patientName, phone, email, city, source, serviceInterested are required fields enforced by the NewLeadDialog form.
- Email must contain '@' character (validated in Login module pattern, applied consistently).
- patientName, phone, city are used as a primary duplicate detection key — exact string match.
- strictAssign requires lastAgentId to be present; if absent, the lead is returned without agent override.

Workflow

1. BDM opens the Leads page and clicks 'New Lead'.
2. NewLeadDialog renders with an empty form.
3. User enters all required fields including createdBy designation.
4. On submit, handleAddLead() is invoked.
5. resolveAgent() checks strictAssign flag and applies last-agent override if applicable.
6. isDuplicateLead() runs against all existing leads (see Duplicate Detection Module).
7. If no duplicates, addLead() commits the lead to LeadsContext state.
8. A 'created' ActivityEntry is prepended to the lead's activityLog.
9. The dialog closes and the new lead appears in the Leads table.

Dependencies

- NewLeadDialog component (UI form)
- LeadsContext (useLeads hook for addLead, addPatientOnly)
- AuthContext (useAuth for role-based field visibility)
- isDuplicateLead() utility function
- resolveAgent() utility function

Section A: Duplicate Detection Module

Description

The Duplicate Detection Module intercepts every new lead creation attempt and performs a multi-field scoring comparison against all existing leads. If a match score of 3 or higher is detected, the system halts the creation flow and presents a DuplicateAlertDialog allowing the user to review matches, merge into an existing record, or proceed with a new entry.

Functional Overview

1. Duplicate detection is handled by a function.
2. The function calculates a **match score using five fields (Phone, Patient Name, City Patient Age, Medical Problem)**
3. The **DuplicateAlertDialog** displays a **detailed comparison card** for matched leads.
4. The system highlights **which fields caused the duplicate match**.
5. Users are given **two options**:
 - **Merge** the lead
 - **Proceed Anyway** and continue creating the lead.

Fields Specification

Field Name	Data Type	Required	Description
phone	Number	Yes	Primary duplicate key — exact string match scores 1 point
patientName	String	Yes	Case-insensitive name match scores 1 point
city	id (Select Dropdown)	No	Case-insensitive city match scores 1 point
patientAge	Number	No	Exact age match scores 1 point
medicalProblem	String	No	Case-insensitive condition match scores 1 point
matchedFields	Array<String>	N/A	UI-rendered list of fields that triggered the match for each candidate

Business Rules

- A lead is flagged as duplicate if `isDuplicateLead()` returns `score >= 3`.
- The comparison is performed against ALL existing leads in the `LeadsContext` store.
- Multiple duplicate matches are supported — all are shown in the dialog simultaneously.
- 'Merge' action redirects the user to the existing lead's detail page (`/leads/:id`) and discards the pending creation.
- 'Proceed Anyway' bypasses duplicate detection and creates the lead without merging.
- 'Cancel' dismisses the dialog without creating or merging any record.
- Merging does not delete any data — it only navigates to the existing record. Actual data consolidation is a manual process.
- The dialog displays Service, Status, Agent, and Consultation status of each matched lead for comparison context.

Validation Rules

- Score threshold is hardcoded at 3 — matching 3 or more of the 5 tracked fields triggers the alert.
- String comparisons are case-insensitive (`.toLowerCase()`).
- Empty or null field values are excluded from scoring (falsy check applied to both incoming and existing records).

Workflow

1. User submits new lead form via `handleAddLead()`.
2. System calls `isDuplicateLead(newLead, existingLead)` for every lead in the store.
3. Matches (`score >= 3`) are collected into `duplicateMatches` array.
4. If `duplicateMatches.length > 0`, `DuplicateAlertDialog` is shown with all matches.
5. Each match card displays patient initials, name, phone, city, lead ID, and matched field badges.
6. User selects: Merge → navigate to existing lead | Proceed Anyway → `addLead()` | Cancel → abort.

Section B: Lead Filtering & Search

Description

The Lead Filtering Module provides a multi-dimensional filter bar on the Leads page that allows users to narrow the visible lead list by search text, patient name, lead status, source channel, assigned agent, relationship type, enquiry type, purpose of call, and case type. Filters combine using AND logic.

Functional Overview

1. All filtering is performed **client-side** using **React useMemo**.
2. A **single consolidated filter predicate** is used for filtering.
3. **Status overrides** from inline status changes are updated and reflected **in real-time**.
4. **Agents have a limited filter view**, meaning the **agent filter is not available** for them.
5. The **Export button is visible** in the interface.
6. The **Export functionality is currently a UI placeholder** and not fully implemented.

Fields Specification

Filter Field	Default	Description
q (search)	""	Free-text search across lead name, patient name, and phone number
patientQ	"all"	Filter by specific patient name; options populated from all unique patient names
status	"all"	New Active Converted Lost Affordable Rehab Interested Follow-up Not Interested
source	"all"	Facebook Instagram Website Referral
agent	"all"	Filter by assigned agent ID (hidden for Agent role)
relationship	"all"	Self Mother Father Spouse Friend
enquiryType	"all"	Promotion Jobs Collaboration Telecalling Rehab Follow-up Patient Enquiry
purposeOfCall	"all"	Promotion Jobs Collaboration Telecalling Rehab Follow-up Patient Enquiry General Inquiry
caseTypeFilter	"all"	Addiction Custom Wellness Detox Others

Business Rules

- All filters are combined with AND logic — a lead must satisfy every active filter to appear.
- patientOnlyIds are always excluded from the filtered list regardless of filter state.
- Status filtering respects statusOverrides — in-session status changes are reflected immediately.
- Agent role (BDM Team Member / Agent) does not see the Agent filter dropdown.
- The patient dropdown is sorted by frequency — patients with more leads appear first.
- Search (q) checks name (l.name or callerName), patientName, and phone simultaneously.
- Filtered count is displayed in the PageHeader: 'X of Y leads · Z unique patients'.

Workflow

1. User modifies any filter control.
2. React state updates trigger useMemo recomputation of the filtered array.
3. Table re-renders showing only matching leads.
4. Filter count in header updates to reflect current match count.

Dependencies

- useLeads() context for the leads array and patientOnlyIds
- agents data for the agent filter dropdown
- AuthContext for role-based filter visibility

Section C: Lead Detail

Description

The Lead Detail Module presents a comprehensive single-lead view including extended editable fields, a multi-patient selector, a lead lifecycle timeline, an activity timeline, session history, transaction history, and patient feedback. The module supports inline editing of all extended fields through an edit mode toggle.

Functional Overview

1. **LeadDetail** retrieves data from both **static leads data (getLead)** and **live data (useLeads / LeadsContext)**.
2. The page contains a **top information card** with **extended lead details**.
3. **Workflow banners** are displayed based on the **service type**.
4. A **cancellation stepper** is shown for **cancelled consultations**.
5. The page uses a **two-column layout** for better organization.
6. One column displays the **timeline section**.
7. The other column contains **side panels with additional information**.

Lead Lifecycle Event Types

Event Type	Icon	Color
Call Received	PhoneIncoming	bg-info/15 text-info
Lead Created	ClipboardList	bg-muted text-muted-foreground
Transferred to Agent	UserPlus	bg-purple-500/15 text-purple-700
Follow-up Scheduled	Clock	bg-warning/20 text-warning
Transferred for Follow-up	ArrowLeftRight	bg-purple-500/15 text-purple-700
Consultation Scheduled	Calendar	bg-success/15 text-success
Consultation Cancelled	XCircle	bg-destructive/15 text-destructive
Payment Completed	CreditCard	bg-success/15 text-success
Refund Initiated	Undo2	bg-warning/20 text-warning
Session Started	PlayCircle	bg-teal-500/15 text-teal-700
Session Completed	CheckCircle2	bg-success/15 text-success
Feedback Submitted	Star	bg-warning/20 text-warning

Business Rules

- Cancellation Stepper shows when consultationStatus === 'Cancelled' — 4 stages: Scheduled → Cancelled → Refund Initiated → Refunded.
- Stage is computed: paymentStatus === 'Refunded' → stage 3; has 'Refund Initiated' history event → stage 2; consultationStatus === 'Cancelled' → stage 1; else stage 0.
- Edit mode is toggled by a button that switches between Edit and Save labels. Cancel button is available in edit mode.
- handleSaveEdit() generates a 'note' ActivityEntry if notes content has changed.
- handleFollowUpSave() generates a 'follow_up' ActivityEntry when a follow-up date is saved.
- Patient selector allows switching between all patients associated with the same callerName.

- Session history shows notes from callerLeads filtered by selectedPatient, sorted newest first.
- serviceType = 'Online Session' renders an info banner with 'Book Session' CTA.
- serviceType = 'Rehab' renders a warning banner with 'Push to PCMS' CTA.
- strictAssign flag renders a STRICT badge (⚡) next to the caller name.

Workflow

1. User navigates to /leads/:id.
2. LeadDetail fetches lead from getLead(id) with LeadsContext fallback.
3. Top card displays all lead info. Extended fields shown read-only.
4. User clicks Edit button — editMode = true, fields become interactive.
5. User modifies fields (status, enquiryType, serviceType, notes, etc.).
6. Status change via dropdown auto-logs a status_change ActivityEntry.
7. Follow-up date change auto-logs a follow_up ActivityEntry.
8. User clicks Save — handleSaveEdit() runs, notes changes logged, editMode = false.
9. Workflow banners update based on serviceType.
10. Cancellation Stepper renders if consultationStatus = Cancelled.

Section D: Call History & AI Insights Module

Description

The Call History module renders a side Sheet from the Leads table displaying all calls associated with a caller (matching callerName across all leads). Each call row is clickable to reveal AI-generated insights in a panel below the table. The module also shows an Activity Timeline for the lead within the same sheet.

Fields Specification — Call Record

Field Name	Data Type	Required	Description
id	id	Yes	Unique call identifier
leadId	id (LeadID)	Yes	Associated lead ID
agentId	id (AgentID)	Yes	Agent who handled the call
timestamp	ISO DateTime	Yes	Date and time of the call
duration	Number (seconds)	Yes	Call duration in seconds

Field Name	Data Type	Required	Description
critical	Boolean	No	Flags calls with destructive background highlight in table
aiInsights.summary	String	Yes	AI-generated call summary
aiInsights.sentiment	Id (Select Dropdown)	Yes	Call sentiment: positive negative neutral
aiInsights.intent	String	Yes	Caller's primary intent (e.g., 'consultation inquiry')
aiInsights.outcome	id (Select Dropdown)	Yes	Call outcome: Converted Follow-up Callback No Action
aiInsights.leadScore	Number (0–10)	Yes	AI lead quality score for this specific call
aiInsights.improvementNotes	String	Yes	AI coaching note for the agent
aiInsights.nextAction	String	Yes	Recommended next step from AI analysis
aiInsights.agentScores.empathy	Number (0–10)	Yes	AI score for agent empathy in this call
aiInsights.agentScores.professionalism	Number (0–10)	Yes	AI score for agent professionalism
aiInsights.agentScores.conversionSkill	Number (0–10)	Yes	AI score for agent conversion skill

Business Rules

- leadCalls is filtered to calls where leadId is in callerLeadIds — all leads sharing the same callerName.
- Calls are sorted newest-first (descending timestamp).
- activeSheetCall drives the AI Insights panel — only shown when a row is clicked.
- Sentiment value is mapped to StatusChip: positive → 'Excellent', negative → 'Critical', neutral → 'Good'.
- Critical flag renders the row with a destructive/5 background.

MODULE 3: Patient Module

Description

The Patient Module provides a centralized patient management interface that groups all lead records under unique patients. It allows users to search, expand, and review patient history, consultation details, payment status, assigned psychologists, BDM information, and complete call history with AI-generated insights.

Functional Overview

1. The module **groups leads based on patientName**.
2. A **unique patient list** is displayed in the system.
3. The module supports **search functionality** for patients.
4. **Expandable detail views** are available for each patient record.
5. Each patient record includes:
 - **Consultation status**
 - **Payment status**
 - **Related leads**
 - **Assigned staff details**
 - **AI-powered call analytics**
6. Users can **expand patient rows** to view more details.
7. Expanded details include **lead-level consultation history** and **communication history**.

Fields Specification

Field Name	Data Type	Required	Description
patientName	String	Yes	Full name of the patient
phone	Number — phone number stored as Number	Yes	Patient contact number
callerName	String	Yes	Name of the caller associated with the patient
relationship	id (Select Dropdown)	Yes	Relationship between caller and patient
leads	Array of UUID (String) — collection of Lead record IDs linked to this patient	Yes	Collection of all lead records linked to the patient

Field Name	Data Type	Required	Description
serviceType	id — user-entered value from Service Type Master form	Yes	Delivery mode: Online Session Rehab In-Person Other
consultationStatus	id — user-entered value from Consultation Status Master form	Yes	Consultation state: Scheduled Completed Cancelled
psychologistAssigned	UUID (String) — references User Master (Psychologist role)	No	Assigned psychologist ID
assignedBDM	UUID (String) — references User Master (BDM role)	No	Assigned BDM ID

Business Rules

- Patients are grouped using patientName as the unique aggregation key.
- Search supports patient name, caller name, and phone number filtering.
- Patient rows are expandable and collapsible using toggleExpand().
- Latest consultation and payment statuses are displayed from the most recent lead.
- Call history is grouped by leadId using callsByLead mapping.
- If no calls exist for a lead, the UI displays 'No calls recorded for this lead.'
- AI-generated insights include sentiment, outcome, and call summaries.
- Call durations are displayed in minutes and seconds format.
- Psychologist and BDM information are resolved dynamically from their respective datasets.
- Results are sorted alphabetically by patient name.

Workflow

1. User navigates to the Patients page.
2. System aggregates all leads into unique patient records.
3. KPI summary cards display total, scheduled, completed, and cancelled patients.
4. User searches using patient name, caller name, or phone number.
5. Matching patient records are displayed in the table.
6. User clicks a patient row to expand detailed lead history.

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7. System displays lead information including psychologist, BDM, consultation status, and payment status.
 8. Associated call history is loaded for each lead.
 9. AI-generated call insights are displayed including sentiment, outcome, and summary.
 10. User collapses the patient row to return to summary view.

MODULE 4: Consultation Booking Module

Description

The Consultation Booking Module allows BDM team members and psychologists to schedule new consultation sessions. It provides a calendar date picker, a time slot grid showing real-time availability per psychologist, a patient search combobox, and a service type selector. For psychologists, the booking is automatically pre-filled to their own account.

Functional Overview

1. Available slots are calculated using a **bookedMap**.
2. The **bookedMap** includes all **active (non-cancelled) consultations**.
3. The system also considers **scheduledSlots fixtures** while computing slot availability.
4. Psychologists can **block slots manually** through the **BlockedSlotsContext**.
5. Blocked slots are marked as **unavailable for booking**.
6. The booking dialog requires a **time slot to be selected** before confirmation.
7. For **non-psychologist roles**, selecting a **psychologist is mandatory** before confirming the booking.

Fields Specification

Field Name	Data Type	Required	Description
bookDate	Date	Yes	Selected calendar date for the consultation
bookSlot	String (HH:00)	Yes	Selected hourly time slot from the 09:00–18:00 grid
patientId	id — free-text input to filter patient list; not a stored field	No	Free-text input to filter patient list by name or phone
serviceType	id — user-entered value from Therapy Service Master form (e.g. Child Therapy, Depression Counseling)	Yes	Service to be delivered: Child Therapy Depression Counseling Marriage Counseling Anxiety Therapy Family Therapy Trauma Recovery Stress Management
date	ISO Date	Yes	Combined bookDate + bookSlot for the consultation record
time	ISO Time	Yes	Combined BookTime + bookSlot for the consultation record
status	id — auto-set to Scheduled on consultation creation; references	Yes	Initial status on creation: Scheduled

Field Name	Data Type	Required	Description
	Consultation Status Master		
psychId	UUID (String) — references User Master (Psychologist role)	Yes	Psychologist ID stored on the consultation record

Transaction History Fields

Field Name	Data Type	Required	Description
id	String	Yes	Unique transaction ID (e.g., TXN-001)
date	ISO DateTime	Yes	Date and time the transaction was initiated
amount	Number	Yes	Transaction amount in Indian Rupees (INR)
paymentMethod	String	Yes	Payment method used (e.g., UPI, Card, Net Banking)
status	Id (Select Dropdown)	Yes	Transaction outcome: Success Pending Failed
refundStatus	Id (Select Dropdown)	Yes	Refund state: None Initiated Completed

All Available Time Slots

Slot	Display
09:00	9:00 AM
10:00	10:00 AM
11:00	11:00 AM
12:00	12:00 PM
13:00	1:00 PM
14:00	2:00 PM
15:00	3:00 PM
16:00	4:00 PM
17:00	5:00 PM
18:00	6:00 PM

Business Rules

- Slot availability is computed per psychologist: a slot is unavailable if it appears in bookedMap for that psychologist on that date.
- Slots blocked via BlockedSlotsContext render with a Lock icon and 'Unavailable' tooltip showing the block reason.
- Psychologist role has bookPsych auto-set to their own ID (PSYCH_EMAIL_MAP lookup) and the field is locked (cannot be changed).
- BDM/Admin roles see a 2-column grid of all psychologists with a workload percentage bar for each.
- A psychologist is shown as unavailable in the grid if the selected time slot is already booked or blocked for them.
- Cancelled consultations are excluded from the bookedMap computation.
- Patient search searches by patientName (case-insensitive) or phone number.
- Confirm Booking is disabled if no slot is selected or (for non-psychologist) no psychologist is selected.

Workflow

10. User clicks 'Book Consultation' button.
11. Dialog resets with today's date selected.
12. User selects a date from the calendar.
13. Slot grid renders with availability computed for the selected date.
14. User selects an available slot.
15. User searches for and selects a patient.
16. User selects service type.
17. BDM/Admin selects a psychologist from the grid; Psychologist role is pre-set.
18. User clicks Confirm Booking — consultation record is created.

Section A: Consultation Room

Description

The Consultation Room Module provides a live video consultation interface for psychologists. It features a three-panel layout with a patient info sidebar, a simulated video canvas, and a session notes panel. On session end, a comprehensive feedback dialog captures clinical notes, patient assessment, assessment file attachments (sent via WATI), risk assessment, patient consent, and recommended follow-up sessions.

Functional Overview

1. The consultation room is accessed through **/consultation-room/:leadId**.
2. An optional **?from= parameter** controls the return navigation path.
3. A **live timer** tracks the consultation session duration.
4. The standard session duration is **45 minutes**.
5. If the session ends before **45 minutes**, an **early-end reason is mandatory**.
6. Users can **upload assessment files** during the session.
7. Assessment files can be **sent to patients through a WATI WhatsApp deep-link**.

Fields Specification

Field Name	Data Type	Required	Description
seconds	Number	Yes	Live session duration counter in seconds (increments every 1s)
mic	Boolean	Yes	Microphone toggle state — true = active
cam	Boolean	Yes	Camera toggle state — true = active
Screen Share	Boolean	Yes	Screen share toggle state — true = sharing
notes	String	No	Live session notes in the right panel — auto-saved indicator (800ms debounce)
savedAt	Number (timestamp)	Yes	Last save timestamp for the 'Saved Xs ago' indicator
feedback	String	No	Patient assessment text shared with the care team
followUpCount	Number	No	Number of recommended follow-up sessions: 0 1–10
recommendations	String	No	Additional clinical recommendations text
earlyEndReason	String	Conditional	Required if session ends before SCHEDULED_DURATION_SECS (2700s = 45min)
endedEarly	Boolean (computed)	N/A	seconds < SCHEDULED_DURATION_SECS
SCHEDULED_DURATION_MINS	Constant: 45	N/A	Standard session duration in minutes

Assessment File Fields

Field Name	Data Type	Required	Description
UUID (String) — system auto-generated on file upload	id (UUID)	Yes	Unique identifier for the uploaded file
name	String	Yes	Original filename
size	Number (bytes)	Yes	File size in bytes
type	String (MIME)	Yes	MIME type (audio/*, video/*, image/*, application/*)
dataUrl	String (base64)	Yes	FileReader base64 data URL for preview
sentFiles	Set<fileId>	N/A	Set of file IDs that have been sent via WATI

Business Rules

- earlyEndReason is mandatory when endedEarly === true. The Submit button is disabled (earlyReasonValid === false) until this field is populated.
- SCHEDULED_DURATION_MINS is hardcoded at 45. Sessions shorter than 45 minutes trigger the early-end warning banner.
- File uploads accept: audio/*, video/*, image/*, .pdf, .doc, .docx, .txt, .ppt, .pptx, .xls, .xlsx.
- WATI integration: sendFileViaWati() opens a WATI deep-link (app.wati.io) in a new tab pre-filled with the patient's phone and a templated message.
- Falls back to wa.me URL format via buildWhatsAppUrl() for environments without WATI.
- sendAllViaWati() sends a single message listing all uploaded files to the patient.
- On successful submission, a toast notification is shown with the assigned agent name and the session is returned to the returnPath.
- The 'Saved Xs ago' indicator updates whenever notes content changes, with a 800ms debounce.
- Feedback and session notes are sent to the assignedAgent and BDM Team Leader on submit.

Workflow

- Psychologist navigates to /consultation-room/:leadId (from Consultations or Schedule page).
- Timer starts automatically.
- Psychologist conducts session — toggles mic/cam/screen as needed.
- Session notes are typed in the right panel and auto-save indicator updates.
- Psychologist clicks 'End' button — openEndDialog() resets all end-session state.
- If endedEarly: warning banner shows; earlyEndReason textarea is required.

- Psychologist fills feedback, uploads assessment files, sends via WATI.
- Selects follow-up session count and enters recommendations.
- Clicks 'Submit & End Session' — handleSubmitAndEnd() validates earlyReasonValid.
- Toast shown; navigation to returnPath.

A.1 Backend Automation — Session End Workflow

Description

When a psychologist completes a consultation session and clicks **Submit & End Session**, the backend system automatically processes all post-session activities. This includes validation of mandatory session details, storage of consultation records, generation of patient feedback links, and automatic delivery of selected assessment documents through **WATI WhatsApp integration**.

The automation ensures that patients receive required post-session materials without manual intervention while maintaining consultation tracking, delivery status monitoring, and audit logs.

Backend Functional Workflow

1. Psychologist completes the consultation and clicks **Submit & End Session**.
2. Backend validates mandatory session fields:
 - Session feedback
 - Psychologist assessment details
 - Early end reason (mandatory if session duration is less than **45 minutes**)
 - Follow-up recommendation details
 - Risk assessment details
 - Patient consent confirmation
3. Backend saves consultation session data in the database.
4. Selected assessment documents uploaded during the consultation are validated and collected.
5. Backend generates a secure patient feedback link for the completed consultation.
6. System prepares WhatsApp message content including:
 - Assessment documents
 - Session completion summary
 - Patient feedback submission link
7. Backend automatically triggers **WATI WhatsApp API**.
8. Patient receives:
 - Selected assessment documents
 - Session completion message
 - Feedback submission link
9. Consultation status is automatically updated to **Completed**.
10. Session completion activity log is created.
11. WhatsApp delivery status is recorded in the system logs.

Backend API Function

Function Name: handleSessionEnd()

Responsibilities

- Save completed consultation session details
- Store psychologist notes and patient assessment
- Validate and upload selected assessment documents
- Generate secure patient feedback URL
- Automatically trigger WATI WhatsApp message
- Update consultation status to **Completed**
- Create activity log entry (**Session Completed**)
- Store WhatsApp delivery status
- Trigger retry mechanism if message delivery fails

WhatsApp Message Template

Hello {{patientName}},

Your consultation session has been completed successfully.

Please find the selected assessment documents attached for your reference.

Kindly share your valuable feedback using the link below:

{{feedbackLink}}

Thank you,
Abhasa Care Team

Database Fields

Field Name	Data Type	Description
feedbackLink	String	Auto-generated patient feedback URL
assessmentFiles	Array	Selected assessment documents uploaded during session
whatsappSentAt	DateTime	Timestamp when WhatsApp message was sent

sessionCompletedAt	DateTime	Session completion timestamp
consultationStatus	Id — auto-updated to Completed after successful session submission; references Consultation Status Master	Updated consultation state: Completed

Business Rules

- Session data must be saved successfully before WhatsApp notification is triggered.
- Only psychologist-selected assessment documents shall be sent to the patient.
- Feedback link must be generated uniquely for each consultation session.
- WhatsApp message shall trigger automatically after successful session completion.
- If consultation duration is below **45 minutes**, early end reason becomes mandatory.
- Consultation status shall automatically change to **Completed** after successful submission.
- A **Session Completed** activity log entry must be created automatically.
- WhatsApp delivery status must be stored for audit tracking.
- If WATI message delivery fails, an automatic retry mechanism shall be triggered.
- Backend must prevent duplicate message sending for the same consultation session.

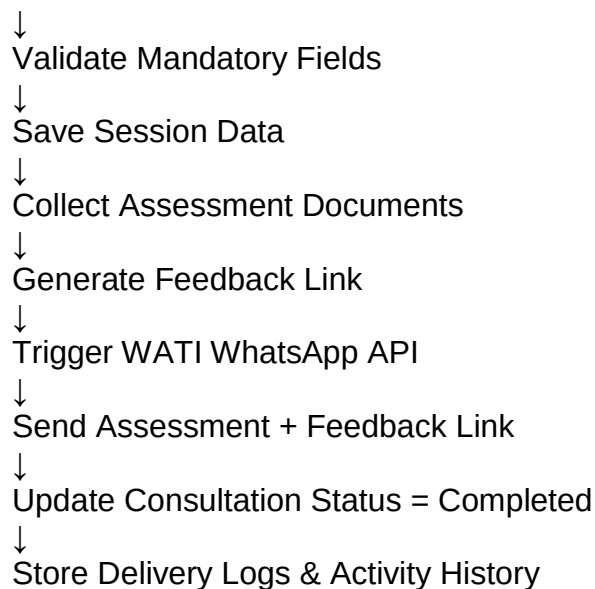
Validation Rules

- **Session feedback** is mandatory.
- **Assessment details** are mandatory.
- **Early end reason** is required when session duration < **45 minutes**.
- **Patient consent confirmation** must be provided before submission.
- Only supported document types are allowed:
 - PDF
 - DOC
 - DOCX
 - TXT
 - PPT
 - PPTX
 - XLS
 - XLSX
 - Images

-
- Audio/Video files
-

Workflow Diagram

Submit & End Session



Section B: Consultation Schedule

Description

The Consultation Schedule Module provides a weekly calendar grid view for psychologists to manage their consultation schedule. Features include: week navigation, today's schedule summary table, slot blocking with mandatory reason, slot unblocking, session start/end from calendar view, and a detail sheet for each consultation. The end-session dialog is feature-equivalent to the Consultation Room.

Functional Overview

1. The calendar displays a **7-column layout (Monday to Sunday)**.
2. It includes **10 time-slot rows (09:00 AM to 06:00 PM)**.
3. A cell shows a **consultation card** if a session is scheduled.
4. A cell shows an **'Unavailable' button** if the slot is blocked.
5. Empty slots provide a **hover-to-block option**.
6. The **BlockedSlotsContext** manages blocked slot states across the application.
7. Clicking a calendar cell opens a **detail sheet from the right side**.

Fields Specification

Field Name	Data Type	Required	Description
currentWeek	Date	Yes	Reference date for the displayed week (Mon–Sun range)
calendarDate	Date	Yes	Currently selected date in the mini calendar picker
Physcholosist id	UUID (String) — references Consultation record	No	ID of the consultation whose detail Sheet is open
overrides	Record<consultId, ConsultStatus>	N/A	Session-scoped status overrides
pendingBlock.dayKey	String (yyyy-MM-dd)	No	Date of the slot being blocked
pendingBlock.hour	Number	No	Hour of the slot being blocked (9–18)
blockReason	String	Conditional	Required reason text for blocking a slot
pendingRemove.dayKey	String	No	Date of the block being removed
pendingRemove.hour	Number	No	Hour of the block being removed
pendingRemove.reason	String	No	Original block reason shown in unblock confirmation
available	string	String	N/A
unavailable	string	String	N/A
status	id		
sessionRecords	Record<consultId, SessionRecord>	N/A	Completed session data keyed by consultation ID

SessionRecord Fields

Field Name	Data Type	Required	Description
psychNotes	String	No	Private clinical notes from the session
earlyEndReason	String	Conditional	Required if session ended before scheduled duration
actualSecs	Number	Yes	Actual session duration in seconds
sessionId	UUID (String) — system auto-generated	Yes	Unique identifier for this session record
endedEarly	Boolean	Yes	Whether session was shorter than SCHEDULED_DURATION_MINS (45)
timestamp	ISO DateTime	Yes	When the session was ended
followUpCount	Number (0–10)	Yes	Number of recommended follow-up sessions
recommendations	String	No	Additional clinical recommendations
patientRating	Number (0–5)	No	Star rating submitted by patient via feedback link
patientComment	String	No	Text comment from patient feedback
patientId			
leadId			
feedbackSent	Boolean	Yes	Whether the WATI feedback link was sent to the patient
risk.selfHarm	id — user-entered value from Risk Level Master form (e.g. Low, Medium, High)	Yes	Risk assessment: self-harm level
risk.aggressiveBehavior	id — user-entered value from Risk Level Master form (e.g. Low, Medium, High)	Yes	Risk assessment: aggressive behavior level
risk.substanceAbuse	id — user-entered value from Risk Level Master form (e.g. Low, Medium, High)	Yes	Risk assessment: substance abuse level
consentGiven	Boolean	Yes	Patient verbally confirmed consent to share session data

Business Rules

- Blocking a slot requires a non-empty blockReason — the Confirm button is disabled without it.
- Blocked slots are stored in BlockedSlotsContext (psychId + dayKey + hour + reason).
- A blocked slot prevents it from appearing as available in the Booking dialog.
- Start Session navigates to /consultation-room/:leadId?from=schedule, recording the start time in sessionStartRef.
- End session computes elapsed seconds from sessionStartRef on dialog open.
- Risk assessment has 4 categories (selfHarm, aggressiveBehavior, suicideRisk, substanceAbuse), each with Low/Medium/High values.
- After session completion, a 'Send feedback link to patient' WhatsApp button appears in the detail Sheet.
- buildFeedbackWatiUrl() constructs a deep-link pointing to /feedback/:consultId with a templated WhatsApp message.
- feedbackSent flag is set to true after the feedback link is sent, changing the button label to 'Feedback link sent ✓'.

Workflow — Blocking a Slot

19. Psychologist hovers over an empty grid cell.
20. A 'Ban' icon button appears.
21. User clicks it — requestBlock(dayKey, hour) called.
22. Mark Unavailable dialog opens with date/time displayed.
23. User enters a mandatory reason.
24. Clicks 'Confirm Unavailable' — addBlock() called in BlockedSlotsContext.
25. Cell now shows 'Unavailable' button; slot is locked in booking dialogs.

Workflow — Starting and Ending a Session

26. Psychologist clicks a scheduled consultation card or opens detail Sheet.
27. Clicks 'Start Session' — navigates to Consultation Room.
28. (Alternatively, End Session button in Sheet opens in-schedule end-session dialog.)
29. End Session dialog captures same fields as ConsultationRoom end dialog + Risk Assessment + Consent.
30. On submit, SessionRecord stored in sessionRecords state; consultation status overridden to 'Completed'.
31. Feedback link button appears in detail Sheet.

Section C: Consultations List

Description

The Consultations List Module provides a tabular view of all consultation records with KPI summary cards. It supports cancellation flows (reschedule or refund) and session history expansion (Admin only). Psychologists see a Start button; BDM/Admin roles see a dropdown menu with View Patient and Cancel options.

Functional Overview

1. Consultations are loaded from the **initialConsults data store**.
2. **Local state management** is used to handle consultation status changes.
3. A **bookedMap** is calculated to prevent **double-booking during rescheduling**.
4. The cancellation process follows a **3-step dialog flow**.
5. Step 1: User **chooses an action**.
6. Step 2: User selects **Reschedule or Refund**.
7. Step 3: User **confirms the selected action**.

Fields Specification

Field Name	Data Type	Required	Description
id	id	Yes	Unique consultation identifier
leadId	UUID (String) — references Lead record	Yes	Associated lead/patient ID
datetime	ISO DateTime	Yes	Scheduled date and time for the consultation
status	id — user-entered value from Consultation Status Master form	Yes	Consultation status: Scheduled Completed Cancelled
psychId	UUID (String) — references User Master (Psychologist role)	No	Psychologist assigned to this consultation (may derive from lead)
cancelStep	String — internal UI state value: choose,	N/A	Cancellation flow step: choose reschedule refund

Field Name	Data Type	Required	Description
	reschedule, or refund		
reschedDate	Date	No	New date selected during reschedule flow
reschedSlot	String (HH:00)	No	New time slot selected during reschedule flow
reschedPsy	UUID (String) — references User Master (Psychologist role)	No	Psychologist selected during reschedule flow
refundLink	String (URL)	No	Generated Razorpay refund URL: https://rzp.io/i/:orderId?amount=...

Cancellation Business Rules

- Cancel option is disabled for consultations with status = 'Cancelled' or 'Completed'.
- Reschedule flow: select psychologist + date + slot → `confirmReschedule()` updates status to 'Scheduled' with new datetime.
- Refund flow: `generateRefundLink()` creates a mock Razorpay URL with format <https://rzp.io/i/{randomOrderId}?amount={txnAmount}&ref={consultId}>.
- Default refund amount is ₹4999 if no transaction history is found.
- Refund link can be copied to clipboard or sent via WATI WhatsApp.
- `confirmRefund()` sets the consultation status to 'Cancelled'.
- Session history (expandable row) is visible to Admin role only via the History column button.
- Psychologist role sees only 'Start' action on Scheduled consultations.

Workflow — Cancellation

1. User clicks the '...' menu on a Scheduled consultation row.
2. Selects 'Cancel consultation'.
3. Step 1 dialog: choose Reschedule or Refund.
4. Reschedule path: pick date/psychologist/slot → Confirm Reschedule.
5. Refund path: review patient/amount → Generate Razorpay Link → Copy or Send via WATI → Confirm Cancellation.

Section D: Consultation History

Description

The Consultation History Module provides a searchable, filterable table of all historical consultation records. For Psychologist role, the view is scoped to their own patients. Each row can be expanded to reveal detailed session notes in a timeline card layout. Summary statistics (total patients, completed, sessions, average rating) are shown below the table.

Fields Specification

Field Name	Data Type	Required	Description
statusFilter	String — user-entered value from Consultation Status Master form; or all	No	Status filter: all Scheduled Completed Cancelled
myPsychId	UUID (String) — derived from logged-in user email via PSYCH_EMAIL_MAP; references User Master (Psychologist role)	N/A	Psychologist's own ID derived from PSYCH_EMAIL_MAP[user.email]
totalSessions	Number (computed)	N/A	Sum of all sessionNotes.length across filtered consultations
avgRating	String (computed)	N/A	Average of feedbackHistory[0].rating across consultations with feedback
sessionNote.timestamp	ISO DateTime	Yes	When the session note was recorded
sessionNote.note	String	Yes	Text content of the session note

Business Rules

- Psychologist role: myPsychId is set from PSYCH_EMAIL_MAP; view is filtered to leads where psychologistAssigned === myPsychId.
- Admin/BDM roles see all consultations.
- Session notes expand on row chevron click — disabled (and greyed) if the lead has no session notes.
- Results are sorted by lastActivity descending (newest first).
- avgRating uses only the first feedbackHistory entry per consultation.

MODULE 5: Follow-up Module

Description

The Follow-up Module manages re-engagement activities for leads that have been marked as 'Lost'. It provides a date-filterable, agent-filterable list of follow-up tasks with inline status updates, overdue/due-today detection, and direct navigation to the lead detail page.

Functional Overview

1. The **Follow-up page** filters leads where **status = 'Lost'**.
2. Only leads with a **populated followUpAssignedTo field** are included.
3. **Agents can view only their assigned follow-ups.**
4. **Leaders and Admins can view all follow-up records.**
5. **Date filtering is set to today's date by default.**
6. Users can **update follow-up status inline** using a **Select dropdown**.
7. Status updates can be made **without leaving the page**.

Fields Specification

Field Name	Data Type	Required	Description
followUpStatus	String — user-entered value from Follow-Up Status Master form (e.g. Pending, Called, Not Reachable)	No	Current re-engagement state: Pending Called Not Reachable Converted Rescheduled
followUpDate	ISO DateTime	No	Scheduled date for the follow-up call
leadid	Id		
followUpFeedback	String	No	Agent notes or outcome recorded after the follow-up call is completed
patientid			
followUpAssignedTo	UUID (String) — references User Master (Agent role)	No	ID of the agent responsible for this follow-up
statusOverrides	Record<leadId, FollowUpStatus>	N/A	Session-scoped status overrides (not persisted)

Business Rules

- Only leads with status === 'Lost' AND followUpAssignedTo populated are shown.
- Agents (BDM Team Member / Agent) see only leads where followUpAssignedTo === their agent ID.
- Admin and BDM Team Leaders see all follow-up leads with the agent filter available.

- Default date filter is today (`format(new Date(), 'yyyy-MM-dd')`).
- Date filter uses `isSameDay()` comparison for exact date matching.
- `overdue = isPast(followUpDate) && !isToday(followUpDate)`.
- Overdue leads render a red 'Overdue' badge; today's leads render a yellow 'Today' badge; tomorrow's render a blue 'Tomorrow' badge.
- Status updates are session-local (`statusOverrides` map); not persisted to context.
- Both roles can update the follow-up status inline from the table.

Status Color Reference

Status	Background	Text Color
Pending	bg-muted	text-muted-foreground
Called	bg-info/10	text-info
Not Reachable	bg-destructive/10	text-destructive
Converted	bg-success/10	text-success
Rescheduled	bg-warning/15	text-warning

Workflow

1. User navigates to Follow-up page.
2. All 'Lost' leads with `followUpAssignedTo` populated are loaded.
3. Role filter applied: agents see only their leads.
4. Date filter defaults to today — table shows today's follow-ups.
5. User can search, change date, or filter by agent/status.
6. User clicks the status Select on a row to update the follow-up status.
7. Status override recorded in session state.
8. User clicks 'View →' to navigate to the full lead detail page.

MODULE 6: Weekly Performance Review Module

Description

The Weekly Performance Review Module provides an AI-powered coaching analytics platform for evaluating agent performance by week. It features an overview table with per-agent call volume, conversion rates, and lead score distribution, and an agent detail view with service-wise bar charts, skill radar charts, 4-band lead score analysis, and AI-generated coaching insights per band.

Functional Overview

1. The module supports **year, month, and week selection**.
2. **Week ranges are configurable** based on user selection.
3. Each **agent can be reviewed individually**.
4. Agent calls are divided into **4 lead score bands**.
5. The system generates **AI coaching insights dynamically**.
6. AI insights are based on:
 - **Conversion rate**
 - **Sentiment analysis**
 - **Empathy scores**
 - **Call distribution**
7. The module helps analyze **agent performance and coaching recommendations**.

Fields Specification — Weekly Report

Field Name	Data Type	Required	Description
agentId	UUID (String) — references User Master (Agent role)	Yes	Agent identifier
agentName	String	Yes	Agent's full name
callsHandled	Number	Yes	Total calls handled in the week
leadsConverted	Number	Yes	Leads converted to consultations/patients
consultationBookings	Number	Yes	Consultation sessions booked
revenueContribution	Number	Yes	Revenue attributed to the agent in INR
avgSentimentScore	Number (0–1)	Yes	Average AI sentiment score across all calls
performanceGrade	String — AI-computed performance grade label	Yes	Computed grade label displayed as a StatusChip

Field Name	Data Type	Required	Description
serviceBreakdown	Array<{service, calls, conversions}>	Yes	Per-service call and conversion counts for the bar chart

Lead Score Bands

Band Key	Label	Score Range	Fill Color
0-3	Very Low	0 – 3	hsl(var(--destructive))
3-5	Cold	3 – 5	hsl(var(--warning))
5-8	Warm	5 – 8	hsl(var(--primary))
8-10	Hot	8 – 10	hsl(var(--success))

AI Insights Logic

- `bandAISuggestions()` generates Strengths, Improvements, and Actions per band based on: conversion rate, empathy score (avg), negative sentiment call count, and call percentage.
- Very Low band (0–3): targets qualification improvement; recommends capping cold calls at 20% of schedule.
- Cold band (3–5): targets follow-up within 24 hours; recommends objection response cards.
- Warm band (5–8): targets 30%+ conversion; recommends urgency triggers and micro-commitments.
- Hot band (8–10): targets same-day callback; recommends dedicating 30%+ of schedule to hot leads.
- `aiSuggestionsFor()` generates overall coaching brief: strengths, improvements, and next-week actions from all calls combined.

Skill Radar Data Points

Skill	Data Source
Empathy	Average of <code>agentScores.empathy</code> across all week's calls
Professionalism	Average of <code>agentScores.professionalism</code> across all week's calls
Conversion	Average of <code>agentScores.conversionSkill</code> across all week's calls
Sentiment	<code>avgSentimentScore * 10</code> (scaled to 0–10)
Volume	<code>min(callsHandled / 15, 10)</code> (normalized)

Business Rules

- Week ranges: Week 1 = days 1–7, Week 2 = 8–14, Week 3 = 15–21, Week 4 = 22–end of month.
- Year selection spans ± 2 years from the current year.
- `callsByAgentWeek()` filters calls by agent ID and week number.
- BandBar in the overview table uses 3 colors: destructive (cold), warning (mid), success (hot).
- Agent detail view has a 'Back to Overview' breadcrumb button.

MODULE 7: Authentication Module

Description

The Authentication Module provides a role-based login interface for the Abhasa CRM platform. It supports four user roles with pre-configured demo credentials. Role determines the initial landing route and the features/filters visible throughout the application.

Fields Specification

Field Name	Data Type	Required	Description
email	String (email)	Yes	User's email address for authentication
FirstName	String	Yes	User's first name
Last Name	String	Yes	User's last name
Phone number	Id — stored as text to preserve formatting	Yes	User's primary phone number
Joining Date	Date	Yes	User's joining date
Alternative Phone number	Id — stored as text to preserve formatting	No	Alternative contact phone number
DOB	Date	Yes	User's date of birth
Age	Number	Yes	User's age
Gender	Id — user-entered value from Gender Master form (e.g. Male, Female, Other)	Yes	User gender: Male Female Other
password	String	Yes	User's password (minimum 1 character)
role	Id — user-entered value from Role Master form (e.g. Admin, BDM Team Leader, BDM Team Member / Agent, Psychologist)	Yes (set by auth)	User role: Admin BDM Team Leader BDM Team Member / Agent Psychologist

Demo Credentials

Role	Email	Password	Landing Route
Admin	admin@abhasa.com	admin123	/dashboard
BDM Team Leader	leader@abhasa.com	leader123	/dashboard
BDM Team Member / Agent	agent@abhasa.com	agent123	/dashboard
Psychologist	psychologist@abhasa.com	psych123	/consultation-schedule

Business Rules

- Psychologist role navigates to /consultation-schedule on login; all other roles navigate to /dashboard.
- Email must include '@' character — validated before the login API call.
- Both email and password are required — empty inputs show 'Please enter email and password' error.
- Quick Demo Login buttons fill the email and password fields for one-click demo access.
- AuthContext.login() is the async function that validates credentials; errors are caught and displayed.

MODULE 8: Psychologists Module

Description

The Psychologists Module provides a centralised directory of all psychologists registered in the Abhasa CRM. It displays enriched cards for each psychologist showing specialisation, workload, ratings, and session statistics. Clicking any card opens a side Sheet displaying that psychologist's full schedule for today, pulled from both the scheduledSlots and consultations data stores. The module is accessible to Admin and BDM roles.

Functional Overview

- The module loads all psychologists from the psychologists data store.
- Each psychologist record is enriched at runtime with: assigned leads count, scheduled sessions, completed sessions, and cancelled sessions — computed from the leads data store.
- A search bar filters psychologists by name or specialisation.
- Summary KPI cards display: Total Psychologists, Average Workload %, Total Sessions, and Average Rating.
- Psychologist cards are displayed in a responsive grid (1 / 2 / 3 columns).
- Each card shows: initials avatar, name, specialisation, star rating, workload progress bar, and a 4-stat row (Sessions, Scheduled, Completed, Cancelled).
- Workload bar colour changes dynamically: green (below 60%), amber (60-79%), red (80% and above).
- Clicking a card opens a Sheet showing today's schedule for that psychologist.
- Today's schedule is built from both scheduledSlots (psychId-linked) and consultations (filtered by isToday and psychologistAssigned) with duplicates removed.
- Schedule entries are sorted chronologically by datetime.

Fields Specification — Psychologist Record

Field Name	Data Type	Required	Description
id	UUID (String) — system auto-generated	Yes	Unique identifier for the psychologist record
name	String	Yes	Full name of the psychologist including title (e.g., Dr. Ananya Sharma)
specialization	String — user-entered value from Specialisation Master form	Yes	Primary clinical specialisation (e.g., Cognitive Behavioral

Field Name	Data Type	Required	Description
			Therapy, Child Psychology)
avgRating	Number (0-5)	Yes	Average patient rating computed from all feedback entries
workload	Number (0-100)	Yes	Current workload percentage: computed as (active consultations / capacity) x 100
email	String	Yes	Psychologist email address; used for PSYCH_EMAIL_MAP role resolution
firstName	String	Yes	Psychologist's first name
lastName	String	Yes	Psychologist's last name
dateOfBirth	Date (ISO format)	Yes	Psychologist's date of birth; used for age computation and HR records
age	Number	Yes	Psychologist's current age in years; computed from dateOfBirth
lunchTime	String (HH:mm–HH:mm)	Yes	Designated lunch break window (e.g., 13:00–14:00); slot blocked automatically during this period
phone	String — stored as text to preserve formatting	Yes	Primary contact phone number; used for appointment notifications and WATI communications
alternativePhone	String — stored as text to preserve formatting	No	Alternative contact phone number for the psychologist

KPI Summary Cards

Field Name	Data Type	Required	Description
Total Psychologists	Number	N/A	Total count of all psychologists in the system
Avg Workload	String (%)	N/A	Average workload percentage across all psychologists, rounded to nearest integer
Total Sessions	Number	N/A	Sum of totalSessions across all enriched psychologist records
Avg Rating	Number (0-5)	N/A	Average of avgRating across all psychologists; displayed to 1 decimal place

Workload Colour Reference

Field Name	Data Type	Required	Description
Below 60%	bg-success / text-success	N/A	Normal workload — psychologist has sufficient availability
60 to 79%	bg-warning / text-warning	N/A	Moderate workload — approaching capacity
80% and above	bg-destructive / text-destructive	N/A	High workload — at or near full capacity; consider rebalancing

Business Rules

- Enrichment is computed using useMemo — recalculated only when psychologists or leads change.
- Filtering by search query (q) matches against both name and specialization fields (case-insensitive).
- Today's schedule merges entries from scheduledSlots (filtered by psychId and isToday) and consultations (filtered by isToday and psychologistAssigned).
- Duplicate prevention: a consultation entry is skipped if an entry with the same id already exists from scheduledSlots.
- Schedule entries are sorted ascending by datetime (earliest first).
- Past scheduled sessions (datetime before now, status still Scheduled) display as In Progress / Overdue.
- The Sheet opens on card click and closes on outside click or Escape.

- Initials are computed from the psychologist name, excluding the prefix Dr. (e.g., Dr. Ananya Sharma becomes AS).

Workflow

- Admin or BDM user navigates to the Psychologists page.
- System loads psychologists and enriches each record with computed session statistics from leads.
- KPI summary cards display aggregate metrics across all psychologists.
- User optionally enters a search query to filter by name or specialisation.
- Matching psychologist cards are displayed in the grid.
- User clicks a psychologist card — the Sheet opens showing today's schedule.
- System filters scheduledSlots and consultations for today's date and the selected psychologist.
- Schedule entries are merged, deduplicated, and sorted chronologically.
- Each schedule entry renders: time, session type icon, patient name, service, and status badge.
- User closes the Sheet to return to the directory grid.

Dependencies

- psychologists data store (imported from @/data/psychologists)
- leads data store (imported from @/data/leads)
- scheduledSlots, consultations data stores (imported from @/data/consultations)
- isToday, format utilities from date-fns
- Sheet, SheetContent components from shadcn/ui

9. Appendix — Role-Based Access Matrix

Module / Feature	Admin	BDM Leader	Agent	Psychologist
Lead Dashboard	✓ Full	✓ Full	X	X
Leads List	✓ Full	✓ Full	✓ Own leads	X
Lead Detail (Edit)	✓ Full	✓ Full	✓ Limited	X
Follow-up Module	✓ All agents	✓ All agents	✓ Own	X
Consultations List	✓ All + History	✓ All	✓ All	✓ Start only
Consultation Schedule	X	X	X	✓ Full
Consultation Room	X	X	X	✓ Full
Consultation History	✓ All	✓ All	X	✓ Own patients
Consultation Dashboard	✓ Full	✓ Full	X	X
Weekly Review	✓ Full	✓ Full	X	X
Agent Filter in Leads	✓ Visible	✓ Visible	X Hidden	X
Duplicate Detection	✓	✓	✓	X
Book Consultation	✓	✓	✓	✓ Own only
Cancel Consultation	✓	✓	✓	X
Refund Link Generation	✓	✓	✓	X
Slot Blocking	X	X	X	✓ Own slots
Risk Assessment Entry	X	X	X	✓
Export Leads (UI)	✓	✓	✓	X

10. Appendix — Integration Points

Integration	Type	Purpose	Implementation
WATI (WhatsApp)	Deep-link URL	Send assessment files, refund links, and feedback links to patients via WhatsApp	buildWatiUrl(phone, msg) → opens app.wati.io/messages?phone=...&message=... in new tab
Razorpay	URL Generation	Generate refund payment links for cancelled consultations	generateRefundLink() → https://rzp.io/i/{orderId}?amount={amount}&ref={consultId}
PCMS	Button CTA (UI)	Push Rehab patient data to Patient Care Management System	'Push to PCMS' button visible when serviceType = 'Rehab'
WhatsApp Fallback	Deep-link URL	wa.me fallback for environments without WATI login	buildWhatsAppUrl(phone, msg) → https://wa.me/{phone}?text=...
Patient Feedback Portal	URL Generation	Post-session feedback link sent to patient via WhatsApp	buildFeedbackWatiUrl() → /feedback/:consultId embedded in WATI message

11. Appendix — Data Relationships

The following diagram describes the key entity relationships in the Abhasa CRM data model:

Entity	Relationship	Target Entity	Key Field
Lead	has many	Patients (via callerName grouping)	callerName
Lead	belongs to	Agent	assignedAgent → agents.id
Lead	belongs to	Psychologist	psychologistAssigned → psychologists.id
Lead	belongs to	BDM	assignedBDM → bdms.id
Lead	has many	Consultations	lead.id → consultation.leadId
Lead	has many	Calls	lead.id → call.leadId
Lead	has many	Transactions	lead.transactionHistory[]
Lead	has many	SessionNotes	lead.sessionNotes[]
Lead	has many	FeedbackEntries	lead.feedbackHistory[]
Lead	has many	ActivityEntries	lead.activityLog[]
Consultation	belongs to	Lead	consultation.leadId
Consultation	belongs to	Psychologist	consultation.psychId
Call	has one	AllInsights	call.aiInsights
Call	belongs to	Agent	call.agentId
BlockedSlot	belongs to	Psychologist	block.psychId